Back to Source: Open-Set Continual Test-Time Adaptation via Domain Compensation

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Abstract

*Test-Time Adaptation (TTA) aims to mitigate distributional shifts between training and test domains. However, existing TTA methods fall short in a realistic scenario where models face both continually changing domains and simultaneous emergence of unknown semantic classes — a challenging setting we term Open-set Continual Test-Time Adaptation (OCTTA). The coupling of domain and semantic shifts often collapses the feature space, severely degrading both classification and out-of-distribution detection. To tackle this, we propose DOmain COmpensation (**DOCO**), an effective framework that robustly performs domain adaptation and OOD detection in a synergistic closed loop. DOCO first performs dynamic, adaptation-conditioned sample splitting to separate likely ID from OOD samples. Using only the ID samples, it learns a **domain compensation prompt** by aligning feature statistics with the source domain, guided by a structural regularizer that prevents semantic distortion. This learned prompt is then propagated to the OOD samples **within** the same batch, isolating their semantic novelty for reliable detection. Extensive experiments on multiple benchmarks show that DOCO outperforms prior continual and open-set TTA methods, establishing a new state-of-the-art for OCTTA. Code is released at https://github.com/ekyle0522/DOCO.*

1. Introduction

Deploying pre-trained models in the real world invariably confronts domain shift, where the test distribution differs from the source and degrades performance. To mitigate this, test-time adaptation (TTA) adapts a source-trained model on the fly using only unlabeled target data at inference. Beyond early single target domain setting [34], recent studies highlight two realistic axes: continual TTA, which copes with nonstationary streams [24, 29, 36, 44], and open-set TTA, where unknown classes co-occur with shifted known ones [9, 20, 42, 43]. We focus on their intersection and formulate the Open-set Continual Test-Time Adaptation (OCTTA) scenario that stresses both stability and unknown-awareness, as depicted in Fig. 2. For instance, a visual perception system deployed in the wild must adapt not only to changing environments, such as from a clear highway scene to a foggy forest scene (domain shift), but also to unexpected objects appearing in those scenes, such as a deer on the road (semantic shift). Crucially, these novel objects are subject to the **same** domain shifts as objects from the known classes, creating a coupled challenge. Robustly handling these combined shifts is therefore important for reliable real-world visual systems.

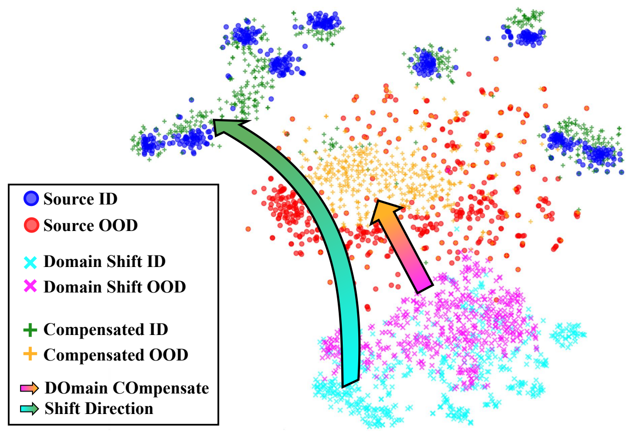

Figure 1. **Illustration of "Back to Source"**. This t-SNE plot shows that while source ID (•) and OOD (•) features are well-separated, a severe domain shift collapses their feature space, mixing ID and OOD features into an inseparable cluster (× and ×). By applying **DO**main **CO**mpensation method, the features of compensated ID (+) and OOD (+) samples are successfully realigned with the original source structure. This disentangles domain and semantic shifts, making corrupted data clearly separable again.

The OCTTA setting poses a tripartite challenge for existing methods. First, the continuous stream of domain shifts exacerbates catastrophic forgetting, eroding knowledge of the source domain as the model adapts to new ones

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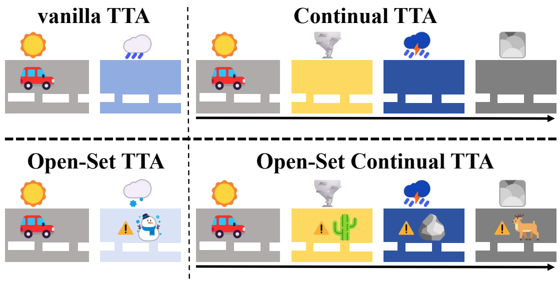


Figure 2. Illustration of different TTA settings.

[36]. Second, the mixture of in-distribution (ID) and out-of-distribution (OOD)[1] samples corrupts the batch statistics required by normalization-based methods and can misguide the optimization in entropy minimization-based approaches [9]. Third, and most critically, is the **antagonistic coupling** of domain and semantic shifts (Fig. 1 and Appendix A.4). A severe domain shift can collapse the feature space, "squashing" the embeddings of both known and unknown classes into a poorly separable region. This collapse blurs class boundaries, simultaneously crippling the model's ability to classify known data and to detect novelties.

To address these challenges, we propose **DO**main **CO**mpensation, a simple and intuitive framework for OCTTA setting. DOCO integrates domain adaptation and OOD detection into a unified, cyclical process. The cycle begins with **Back-to-Source Prompt Learning**, where a lightweight domain compensation prompt is updated using likely ID samples from the current online batch. This update, performed via one-step backpropagation, aligns target feature statistics with those of the source domain to neutralize the domain shift without distorting semantics. This learned prompt, which now encodes the current domain information, is then immediately applied to the likely OOD samples in the same batch through **Intra-Batch Prompt Propagation**. By compensating for their domain shift, this step isolates their semantic novelty, making them more distinguishable from known classes. The loop is sustained and improved by **Adaptation-Conditioned Sample Splitting**: prompt-adapted features enable more accurate ID/OOD separation for the current batch, yielding a purer ID set for prompt learning and better capture of the current domain mode. Improved splitting thus creates a virtuous cycle, enhancing prompts and, in turn, future adaptation and detection. This closed-loop design makes the system resilient to continuous and unforeseen environmental changes.

Our main contributions are as follows:

- We formally introduce Open-set Continual Test-Time Adaptation (OCTTA), a pragmatic setting that reflects real-world complexities. We show that prior TTA methods suffer performance degradation caused by coupled domain and semantic shifts in the feature space.
- We propose DOCO, a visual prompt learning-based framework that effectively mitigates the negative coupling of shifts through domain compensation and establishes a positive feedback loop via a dynamic sample-splitting mechanism.
- We conduct extensive experiments on multiple OOD datasets under the OCTTA setting. The results show that DOCO achieves state-of-the-art performance, especially surpassing the next-best UniEnt by 4.7% on ImageNet-C, validating its effectiveness and robustness.

[1] In this paper, ID refers to data within the source semantic space, OOD refers to data outside it.

2. Related Work

2.1. Test-time Adaptation

TTA [5, 16, 18, 21, 23, 38] adapts a source-trained model online without labeled target data. We summarize methods along the two axes introduced above. For continual TTA, CoTTA [36] stabilizes long-horizon updates via weight/augmentation-averaged targets with stochastic neuron restoration. EATA [29] filters unreliable or redundant samples and regularizes important weights to curb risky updates and forgetting. SAR [30] replaces brittle BN with batch-agnostic norms and employs sharpness-aware reliable entropy to avoid collapse under wild shifts. ViDA [24] introduces lightweight adapters to decouple domain-shared from domain-specific factors. DPCore [44] retains domain knowledge through a dynamic prompt coreset aimed at recurring or short-lived domains. For open-set TTA, wisdom-of-crowds filtering [20] suppresses losses whose confidence decreases after adaptation. UniEnt [9] jointly minimizes entropy on pseudo-csID and maximizes it on pseudo-csOOD with marginal-entropy regularization. STAMP [42] leverages a stable, class-balanced memory with self-weighted entropy. COME [43] regularizes confidence via conservative entropy minimization to enhance open-world stability.

2.2. Out-of-Distribution Detection

OOD detection [7, 31, 37, 46] separates ID from unknown samples without domain shift. Two widely used lines are post-hoc scoring and test-time detection. For post-hoc scoring, MSP [12] thresholds predicted confidence, and energy-based scoring [25] replaces confidence with log-sum-exp energy and often surpasses MSP and MaxLogit [15]. For test-time detection, RTL [8] learns a lightweight linear map from features to OOD scores directly at inference. AUTO [40] performs online optimization with an in–out-aware filter, an ID memory, and a consistency loss. CODA [3] compacts source embeddings with virtual unknowns and disambiguates known from unknown through prototype-guided updates. Recent dictionary-based designs such as OODD [41] maintain a dynamic OOD feature dictionary and calibrate scores via feature–dictionary similarity during testing.

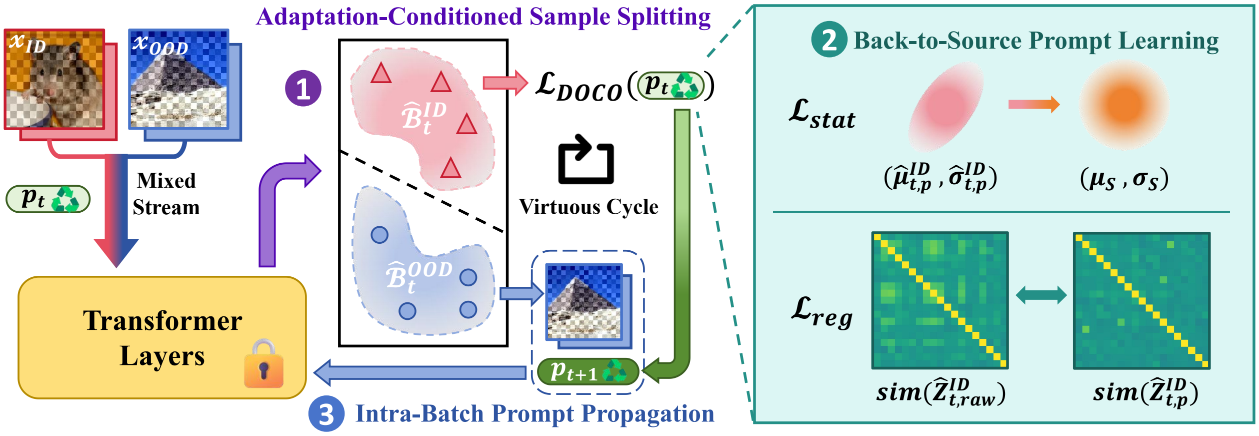


Figure 3. Overview of DOCO framework. ❶ The incoming batch is processed with the prompt p_t to reduce domain shift, and features $z_{i,p}$ are scored against source class prototypes $W_h = \{w_c\}_{c \in \mathcal{Y}^S}$ to split the batch into $\hat{\mathcal{B}}_t^{\text{ID}}$ and $\hat{\mathcal{B}}_t^{\text{OOD}}$. ❷ A new prompt p_{t+1} is optimized on the ID subset by minimizing $\mathcal{L}_{\text{DOCO}}$ in one-step backpropagation. ❸ The newly optimized prompt p_{t+1} is immediately propagated to OOD data within the same batch, neutralizing domain shift for accurate OOD detection.

3. Preliminaries

In this section, we provide a brief overview of Vision Transformers (ViTs) and the Visual Prompt Tuning (VPT) paradigm, and then formally define the OCTTA setting.

ViTs & VPT. We use the ViT-base architecture [6] as our backbone. A model f with parameters θ is decomposed as a feature extractor $\phi : \mathcal{X} \to \mathcal{Z}$ with parameters θ_ϕ and a linear classifier $h : \mathcal{Z} \to \mathbb{R}^C$ with parameters θ_h, i.e., $f_\theta = h \circ \phi$. Concretely, we write $h(z) = W_h z + b_h$, where $W_h = [w_1, \ldots, w_C]^\top$ and $h_c(z)$ denotes the logit for class c. Given an image, we denote by $z \in \mathcal{Z}$ the [CLS] representation after the last transformer block, and obtain predictions via $\hat{y} = \arg\max_{c \in [C]} h_c(z)$. For efficient test-time adaptation, we adopt Visual Prompt Tuning (VPT) [19]. We augment the input sequence with L learnable prompt tokens $p = \{\texttt{[Prompt]}_i\}_{i=0}^{L-1}$, inserted after [CLS] and before the patch tokens, yielding the first-layer input $\{\texttt{[CLS]}, p, \text{patch}_1, \ldots, \text{patch}_k\}$. During adaptation, we freeze the original model parameters θ and update only the prompts p. This parameter-efficient design greatly reduces the number of trainable parameters and mitigates catastrophic forgetting, while still allowing the model to swiftly adjust to shifting test distributions.

OCTTA Problem Formulation. Given a model f_θ pre-trained on a source domain $\mathcal{D}^S = (\mathcal{X}^S, \mathcal{Y}^S)$, the goal is to adapt f_θ to a sequence of unlabeled target domains $\{\mathcal{D}^{T_i}\}_{i=1}^M$. Each target domain $\mathcal{D}^{T_i}$ possesses its label space $\mathcal{Y}^{T_i}$, and the source label space is a proper subset of the target, i.e., $\mathcal{Y}^S \subset \mathcal{Y}^{T_i}$. The model receives an online stream of test batches $\{\mathcal{B}_t\}_{t=1}^\infty$. For any domain $\mathcal{D}^{T_i}$, we model the data distribution $P_{\text{test}}^{T_i}$ with the Huber contamination model [17] to represent coupled domain and semantic shifts:

$$P_{\text{test}}^{T_i} = (1-\kappa) P_{\text{ID-C}}^{T_i} + \kappa P_{\text{OOD-C}}^{T_i}. \tag{1}$$

In this mixture, $\kappa \in [0, 1)$ is a preset ratio for OOD data. While κ controls the overall OOD proportion, each mini-batch is randomly sampled, resulting in a stochastic ID-to-OOD ratio per batch to better simulate real-world data streams. $P_{\text{ID-C}}^{T_i}$ is the distribution of ID samples (x, y) with known class label where $y \in \mathcal{Y}^S$, while $P_{\text{OOD-C}}^{T_i}$ is the distribution of OOD samples with novel classes where $y \in \mathcal{Y}^{T_i} \setminus \mathcal{Y}^S$. The '-C' suffix signifies that both ID and OOD data are affected by the same domain corruption specific to $\mathcal{D}^{T_i}$. This adaptation process unfolds online, where at each time step t, the model's parameters θ_t are updated to θ_{t+1} using only the current batch $\mathcal{B}_t$ and a single backpropagation step. The objective is to correctly classify future samples from known classes $\mathcal{Y}^S$ while detecting those from novel classes $\mathcal{Y}^{T_i} \setminus \mathcal{Y}^S$.

4. Methodology

Overview. We introduce DOCO, a novel framework for the OCTTA setting, with an overview provided in Fig. 3. DOCO consists of three major components and systematically addresses a series of core challenges: (1) How the prompt can adapt to the target domain without semantic overfitting (Sec. 4.1)? (2) How to reuse the learned domain knowledge to bootstrap the model's inference capabilities (Sec. 4.2)? and (3) How to effectively separate ID and OOD samples under severe domain shifts (Sec. 4.3)? These components operate within a synergistic closed-loop, where more accurate splitting enables better prompt learning, which in turn improves inference and guides future adaptation steps. Algorithm is detailed in Appendix A.1.

4.1. Back-to-Source Prompt Learning

Upon receiving a batch $\mathcal{B}_t$ at time t, we first isolate a subset of likely ID samples, denoted as $\hat{\mathcal{B}}_t^{\mathrm{ID}} \subset \mathcal{B}_t$, using a dynamic splitting mechanism detailed in Sec. 4.3. For this subset, we optimize the current prompt p_t and obtain an updated prompt p_{t+1}. The purpose of the prompt is to align the feature distribution of $\hat{\mathcal{B}}_t^{\mathrm{ID}}$ with that of the source domain $\mathcal{D}^S$, thereby counteracting the domain shift while preserving the intrinsic semantic structure of the samples.

Statistical Alignment. A domain shift from $\mathcal{D}^S$ to $\mathcal{D}^{T_i}$ will inevitably cause a shift in the statistics of the feature space [1]. We argue that encouraging the ID feature statistics to move back toward the source statistics enables the prompt to approximate a compensation for the domain shift in latent space [27]. To this end, we pre-cache the mean μ_S and standard deviation σ_S of the source features calculated from a small set of unlabeled samples in an offline manner. For the current ID batch $\hat{\mathcal{B}}_t^{\mathrm{ID}}$, we extract the features using the prompt p_t, yielding $\hat{Z}_{t,p}^{\mathrm{ID}} = \{\phi(x; p_t)\}_{x \in \hat{\mathcal{B}}_t^{\mathrm{ID}}}$. We then compute the batch statistics, mean $\hat{\mu}_{t,p}^{\mathrm{ID}}$ and standard deviation $\hat{\sigma}_{t,p}^{\mathrm{ID}}$. The statistical alignment loss is defined as the L2 distance between the batch and source statistics:

$$\mathcal{L}_{\mathrm{stat}}(p_t) = \|\hat{\mu}_{t,p}^{\mathrm{ID}} - \mu_S\|_2 + \|\hat{\sigma}_{t,p}^{\mathrm{ID}} - \sigma_S\|_2. \tag{2}$$

Structural Preservation. Relying solely on $\mathcal{L}_{\mathrm{stat}}$ is insufficient, as batch statistics reflect both domain shift and batch-specific semantics. For instance, if a batch contains only "dogs" and "cats", forcing its feature statistics to match the entire source statistics with far more classes could compel the prompt to distort the feature structure, causing it to overfit to the batch's narrow semantics rather than learning a general domain compensation.

To address this, we introduce a regularization term to preserve the relative feature geometry. Specifically, we enforce that the pairwise similarity structure within the selected ID subset remains consistent before and after applying the prompt. Let $\hat{\mathcal{B}}_t^{\mathrm{ID}} = \{x_i\}_{i=1}^n$, where $n = |\hat{\mathcal{B}}_t^{\mathrm{ID}}|$. For each x_i, let $z_i^{\mathrm{raw}} = \phi(x_i)$ and $z_i^{p_t} = \phi(x_i; p_t)$ denote its raw and prompted feature representations, respectively. Accordingly, $\hat{Z}_{t,\mathrm{raw}}^{\mathrm{ID}} = \{z_i^{\mathrm{raw}}\}_{i=1}^n$ and $\hat{Z}_{t,p}^{\mathrm{ID}} = \{z_i^{p_t}\}_{i=1}^n$. We use cosine similarity

$$C(a,b) = \frac{a \cdot b}{\|a\|_2 \, \|b\|_2}, \tag{3}$$

which is evaluated on both raw features $(z_i^{\mathrm{raw}}, z_j^{\mathrm{raw}})$ and prompted features $(z_i^{p_t}, z_j^{p_t})$. The structural preservation loss is then defined as the Frobenius norm of the difference between the two pairwise similarity matrices:

$$\mathcal{L}_{\mathrm{reg}}(p_t) = \sqrt{\sum_{i=1}^{n}\sum_{j=1}^{n}\left(C(z_i^{p_t}, z_j^{p_t}) - C(z_i^{\mathrm{raw}}, z_j^{\mathrm{raw}})\right)^2}, \tag{4}$$

or equivalently,

$$\mathcal{L}_{\mathrm{reg}}(p_t) = \left\| \mathrm{sim}(\hat{Z}_{t,p}^{\mathrm{ID}}) - \mathrm{sim}(\hat{Z}_{t,\mathrm{raw}}^{\mathrm{ID}}) \right\|_F, \tag{5}$$

where for a feature set $Z = \{z_i\}_{i=1}^n$, $\mathrm{sim}(Z) \in \mathbb{R}^{n \times n}$ denotes its pairwise cosine-similarity matrix, whose (i,j)-th entry is $C(z_i, z_j)$. By penalizing disruptions to the relative feature geometry, this regularizer encourages the prompt to compensate for domain shift without overfitting to the narrow semantics of the current batch. The final objective for optimizing the current prompt p_t is

$$\mathcal{L}_{\mathrm{DOCO}}(p_t) = \mathcal{L}_{\mathrm{stat}}(p_t) + \beta \mathcal{L}_{\mathrm{reg}}(p_t), \tag{6}$$

where β is a regularization coefficient. The prompt parameters are then optimized on the current ID subset to obtain an updated prompt p_{t+1}.

4.2. Intra-Batch Prompt Propagation

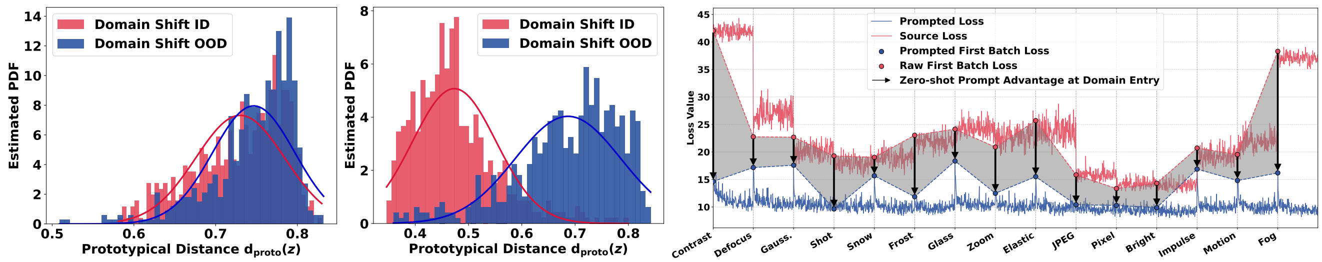
Direct Intra-Batch Knowledge Reuse. Within the current batch, samples in $\hat{\mathcal{B}}_t^{\mathrm{ID}}$ are inferred using the prompt p_t. We then immediately reuse the batch-t domain knowledge learned from these likely ID samples — instantiated as the updated prompt p_{t+1} — by applying it only to likely OOD samples in $\hat{\mathcal{B}}_t^{\mathrm{OOD}}$ within the same batch. Since all samples in $\mathcal{B}_t$ share the same batch-wise domain component δ_t, the prompt p_{t+1} compensates the domain factor of OOD features in a consistent manner. The final prediction for likely OOD samples is produced by the frozen linear classifier head h on compensated features:

$$\hat{y} \;=\; \arg\max_{c \in \mathcal{Y}^S} \; h_c\big(\phi(x; p_{t+1})\big), \qquad x \in \hat{\mathcal{B}}_t^{\mathrm{OOD}}. \tag{7}$$

Motivation. Recent OOD generalization work [28] suggests that domain-invariant semantics can be exposed by removing domain-specific components from representations. As an intuition, we write the batch-t representation as $\phi(x) \approx s(x) + \delta_t$, where $s(x)$ encodes class semantics and δ_t is a batch-wise domain factor. We learn p_{t+1} from only likely ID samples in $\mathcal{B}_t$ and then *immediately* propagate it to the likely OOD subset in the same batch, yielding

$$\phi(x; p_{t+1}) \;\approx\; \phi(x) - \delta_t \;\approx\; s(x). \tag{8}$$

This propagation is non-trivial: it uses ID-only updates to estimate and neutralize the *same* batch factor for all samples sharing δ_t, which (i) corrects mis-split IDs by pulling them back toward source-aligned neighborhoods, (ii) makes true OODs more novel relative to the compensated source geometry, and (iii) avoids leaking pseudo-label noise by not back-propagating through likely OOD samples, thus stabilizing the decision boundary for the whole batch. Comparison between training-time explicit separation [28] and our test-time in-process correction is provided in Appendix A.2.



(a) w/o prompt. (b) w/ prompt. (c) $\mathcal{L}_{\text{stat}}$ across 15 domains w/ and w/o prompt.

Figure 4. (a-b) DOCO facilitates the separation of ID and OOD by disentangling the bimodal distribution of the prototypical distance $d_{\text{proto}}(z)$. (c) For each new domain, the statistical loss $\mathcal{L}_{\text{stat}}(p_t)$ with DOCO is consistently smaller than that of the source model on the first batch, indicating that the learned prompt generalizes well to unseen domains.

4.3. Adaptation-Conditioned Sample Splitting

Domain overshadowing and prompt generalization. The effectiveness of our framework relies on splitting each batch $\mathcal{B}_t$ into likely ID and likely OOD subsets, $\hat{\mathcal{B}}_t^{\text{ID}}$ and $\hat{\mathcal{B}}_t^{\text{OOD}}$. This separation prevents OOD contamination during prompt learning, leading to a cleaner estimation of the batch-wise domain patterns. However, severe domain shifts can overshadow semantic differences (Fig. 4a), causing substantial overlap between the ID and OOD distributions of the prototypical distance d_{proto} (defined in Eq. 9). Therefore, for each batch we first compute compensated features $Z_{t,p} = \{\phi(x; p_t)\}_{x \in \mathcal{B}_t}$ using the prompt p_t. Even under new domains, the structure-preserving prompt exhibits strong cross-domain generalization (Fig. 4c and Appendix A.3), enabling DOCO to address the "Continual" aspect of OCTTA effectively. This "back-to-source" effect restores a clearer bimodality (Fig. 4b), enabling reliable partitioning.

Prototypical Distance Splitting. Given the prompted features $Z_{t,p}$ and the frozen classifier weights $\{w_c\}_{c \in \mathcal{Y}^S}$ serving as source prototypical proxies [10], for each $z \in Z_{t,p}$ we define the *prototypical distance*

$$d_{\text{proto}}(z) = 1 - \max_{c \in \mathcal{Y}^S} C(z, w_c). \tag{9}$$

Here, $C(\cdot, \cdot)$ denotes cosine similarity defined in Eq. 3. Let $\mathcal{S}_t := \{ d_{\text{proto}}(z) \mid z \in Z_{t,p} \}$ denote the collection of prototypical distances induced by this batch. We run K-Means with $K = 2$ over these scalar scores:

$$(\widehat{\mathcal{K}}_0, \widehat{\mathcal{K}}_1) = \arg\min_{\mathcal{K}_0, \mathcal{K}_1} \sum_{i \in \{0,1\}} \sum_{d \in \mathcal{K}_i} \|d - \mu_i\|^2, \quad \text{s.t.} \quad \mathcal{K}_0 \cup \mathcal{K}_1 = \mathcal{S}_t, \;\; \mathcal{K}_0 \cap \mathcal{K}_1 = \varnothing. \tag{10}$$

where $\mu_i = \frac{1}{|\mathcal{K}_i|} \sum_{d \in \mathcal{K}_i} d$ denotes the centroid of cluster $\mathcal{K}_i$. Since a smaller d_{proto} indicates being closer to the source prototypes, we assign the cluster with the smaller centroid to ID. Denoting that index by $i_{\text{ID}} \in \{0, 1\}$ and $i_{\text{OOD}} = 1 - i_{\text{ID}}$, the split is

$$\hat{\mathcal{B}}_t^{\text{ID}} = \{ x \in \mathcal{B}_t \mid d_{\text{proto}}(\phi(x; p_t)) \in \widehat{\mathcal{K}}_{i_{\text{ID}}} \}, \quad \hat{\mathcal{B}}_t^{\text{OOD}} = \{ x \in \mathcal{B}_t \mid d_{\text{proto}}(\phi(x; p_t)) \in \widehat{\mathcal{K}}_{i_{\text{OOD}}} \}. \tag{11}$$

5. Experiments

5.1. Experimental Setup

Datasets. For ID component, we first use the standard benchmark ImageNet-C [11]. This dataset contains 15 common corruption types, and we use the highest severity level 5. To further assess performance, we then adopt a newly-released LAION-C benchmark [22], which produces six challenging, synthetic distortions. Given their significant difficulty, we use a moderate severity level of 3 for these corruptions. For the OOD component, we follow common practices but apply the same corruption to standard OOD benchmarks. Specifically, the OOD datasets are corrupted versions of Places365 [45], Textures [4], NINCO [2], iNaturalist [32], SSB-Hard [33], and SUN [39].

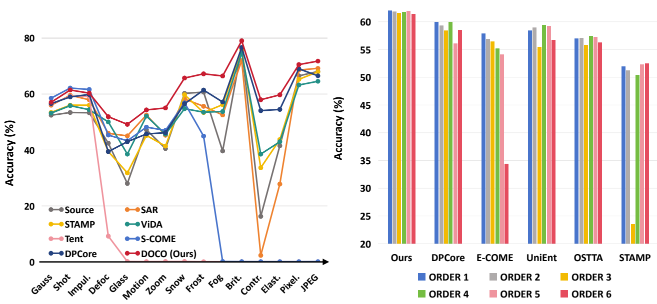
Baselines. We compare our method against strong continual and open-set TTA baselines including Tent [34], CoTTA [36], SAR [30], EATA [29], ViDA [24], DPCore [44], STAMP [42], OSTTA [20], UniEnt [9], and COME [43].

Evaluation Protocol. We use three metrics for a comprehensive evaluation. First, we report ID classification accuracy **(ACC)** to measure domain generalization. Second, we use the area under the ROC curve **(AUC)** to assess the model's threshold-free outlier detection capability. Finally, to jointly evaluate both aspects, we report the **H-score**, which is the harmonic mean of ACC and AUC.

Implementation Details. Experiments use a ViT-B/16 pretrained on ImageNet-1K, with weights from `timm`. For evaluating OOD detection, we use the energy score [25] to compute AUC. The test stream follows Eq. 1 with OOD ratio $\kappa = 0.5$, and we standardize the batch size to 64 for fair comparison. To simulate a real-world deployment and ensure a fair blind test, hyperparameters for all methods are tuned only on the first dataset-domain combination. These settings are then frozen and applied to all subsequent, unseen datasets to rigorously evaluate generalization robustness [42]. Prior to online adaptation, we pre-compute source feature statistics from 300 samples, following EATA

Table 1. Results (%) for ImageNet-to-ImageNet-C (severity = 5, $\kappa = 0.5$) in OCTTA setting across six covariate-shifted OOD datasets. All the results are **averaged over 15** corruptions. **Bold** and <u>underline</u> are used to indicate the first and second best performance, respectively.

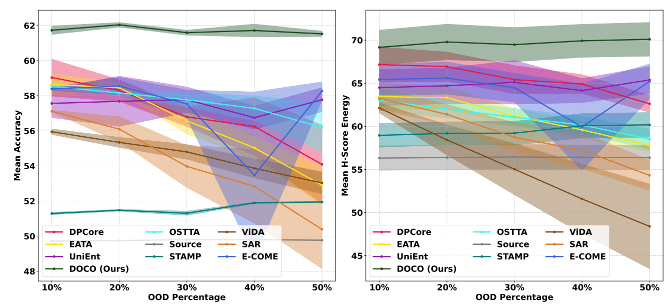
Method	Places.–C		Texture–C		iNatur.–C		SUN–C		SSB-H.–C		NINCO–C		Avg.		
	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	H-score
Source	49.8	66.8	49.8	70.5	49.8	78.7	49.8	71.4	49.8	56.3	49.8	64.5	49.8	68.0	56.4
Tent (ICLR'21) [34]	12.3	49.3	3.8	44.6	12.1	50.1	13.3	38.9	55.3	62.5	37.5	59.8	22.4	50.9	23.8
CoTTA (CVPR'22) [36]	49.9	63.0	48.8	65.5	49.4	75.3	49.6	65.8	49.9	55.5	49.4	62.0	49.5	64.5	54.8
EATA (ICML'22) [29]	54.9	65.4	50.5	72.1	51.9	72.9	52.1	70.8	56.3	58.3	51.8	64.3	52.9	67.3	57.8
SAR (ICLR'23) [30]	51.2	59.6	45.0	60.0	46.0	64.3	49.5	57.9	56.8	60.5	53.7	66.8	50.4	61.5	54.3
OSTTA (ICCV'23) [20]	57.5	60.2	56.5	55.5	50.9	66.3	55.6	59.4	58.8	62.5	57.8	67.2	56.2	61.9	58.5
ViDA (ICLR'24) [24]	53.0	59.8	52.1	36.8	53.1	35.9	51.5	30.4	55.2	61.9	53.2	62.9	53.0	47.9	48.4
UniEnt (CVPR'24) [9]	<u>58.6</u>	74.1	55.5	<u>79.7</u>	56.8	<u>89.6</u>	58.1	<u>84.6</u>	59.1	62.1	<u>58.6</u>	72.2	57.8	<u>77.0</u>	<u>65.4</u>
STAMP (ECCV'24) [42]	51.9	72.4	52.0	75.1	52.1	85.6	51.9	77.4	52.0	60.9	51.9	71.4	51.9	73.8	60.2
E-COME (ICLR'25) [43]	57.7	76.0	<u>57.8</u>	78.6	<u>58.9</u>	85.4	<u>59.3</u>	82.7	59.2	58.9	56.6	71.3	<u>58.3</u>	75.5	65.2
S-COME (ICLR'25) [43]	31.2	58.5	29.3	67.1	22.9	55.6	54.6	76.4	54.3	55.6	53.2	65.1	40.9	63.0	45.5
DPCore (ICML'25) [44]	56.4	<u>76.7</u>	54.3	78.0	46.7	82.8	52.6	82.0	<u>60.2</u>	<u>62.6</u>	54.3	<u>75.3</u>	54.1	76.2	62.6
DOCO (Ours)	**61.8**	**80.6**	**61.0**	**84.6**	**61.4**	**95.7**	**61.5**	**92.2**	**61.9**	**65.9**	**61.5**	**77.4**	**61.5**	**82.7**	**70.1**



(a) Cross domain Acc

(b) Different domain order

Figure 5. Further analysis of the OCTTA setting on the ImageNet-C benchmark. (a) Per-domain accuracy (%) of different models under continuous domain shifts. (b) Accuracy (%) of models tested with different domain orders.



(a) Accuracy (%)

(b) H-score (%)

Figure 6. Performance under different OOD ratios.

[29]. The prompts conduct a one-time self-supervised update for 50 iterations to refine the initial state. During TTA, we update $L = 8$ learnable prompts for each incoming batch using AdamW (learning rate 1e-1) with structural preservation weight $\beta = 0.5$. In contrast to methods like CoTTA which update all model parameters, DOCO exclusively fine-tunes the prompts, while most other baselines update only the affine parameters of LayerNorm layers. For the plug-in frameworks UniEnt and COME, we evaluate their **strongest** open-set and continual configurations: EATA with UniEnt+ (full version, denoted as `UniEnt` for short), EATA with COME (`E-COME`) and SAR with COME (`S-COME`). More details are available in Appendix B.2.

5.2. Main Results

ImageNet-C Benchmark. The comprehensive results in Tab. 1 show that DOCO achieves state-of-the-art performance across multiple OOD datasets. On average[2], our method sets a new SOTA with an H-score of $\mathbf{70.1}\%$, surpassing the next-best method UniEnt by a significant $\mathbf{4.7}\%$. This top-tier result is based on a leading known-class accuracy of $\mathbf{61.5}\%$ and an exceptional OOD detection AUC of $\mathbf{82.7}\%$. This demonstrates DOCO's balanced enhancement of both ID classification and OOD detection. To further analyze DOCO's robustness, we examine its performance across individual domains and different domain orderings in Fig. 5. Figure 5a tracks per-domain accuracy as the model sequentially adapts to 15 corruptions. DOCO's performance curve is consistently at or near the top, showcasing its stable adaptation. This stability is particularly pronounced on difficult domains like *Contrast*, where most other methods suffer a sharp decline in accuracy. DOCO, by effectively decoupling the severe domain shift from the underlying semantic information, remains strong under such challenging conditions. Furthermore, we assess its resilience to the sequence of domains. As depicted in Fig. 5b, DOCO's Accuracy remains stable across six different random domain orderings. This result underscores that DOCO's effectiveness does not depend on a favorable domain sequence, confirming its robustness in truly continual settings. We further investigate how the OOD percentage κ impacts our method. As shown in Fig. 6, we manipu-

[2]ACC/AUC/H-score are first computed per dataset and then averaged.

Table 2. Results (%) for LAION-C benchmark (severity = 3, $\kappa = 0.5$) in the OCTTA setting across six domain shift OOD datasets. All results are averaged over 6 constantly switching domains. '`-L`' stands for applying LAION-C corruption to OOD dataset.

Method	**Places.–L**		**Texture–L**		**iNatur.–L**		**SUN–L**		**SSB-H.–L**		**NINCO–L**		**Avg.**		
	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	H-score
Source	16.2	52.2	16.2	54.8	16.2	58.9	16.2	54.9	16.2	54.8	16.2	55.8	16.2	55.2	19.1
Tent (ICLR'21)	0.2	47.4	0.2	58.9	0.4	46.5	0.2	49.8	0.4	49.3	0.2	53.4	0.3	50.9	0.6
CoTTA (CVPR'22)	16.2	50.7	16.1	51.7	16.1	56.1	16.1	52.8	16.1	54.5	16.1	54.8	16.1	53.4	18.5
EATA (ICML'22)	23.9	56.0	23.3	59.9	20.7	59.6	22.5	60.3	25.0	54.4	23.1	55.1	23.1	57.5	27.9
SAR (ICLR'23)	7.6	50.6	9.4	51.8	4.5	47.9	4.6	48.4	13.6	54.4	5.1	51.5	7.5	50.8	9.6
OSTTA (ICCV'23)	16.3	49.4	16.7	48.6	15.6	47.6	14.5	48.6	17.1	54.4	16.9	53.7	16.2	50.4	17.5
ViDA (ICLR'24)	2.5	42.7	3.2	48.6	2.7	36.5	2.0	40.4	2.2	49.0	2.4	47.1	2.5	44.0	3.9
UniEnt (CVPR'24)	24.0	58.3	23.8	64.5	23.7	66.0	23.6	64.5	24.4	54.4	23.1	57.0	23.8	60.8	29.3
STAMP (ECCV'24)	16.4	53.5	15.6	54.1	15.3	56.5	16.0	53.8	16.2	55.0	15.7	55.9	15.9	54.8	19.5
E-COME (ICLR'25)	4.6	53.1	17.9	57.6	16.5	63.6	18.1	61.4	18.8	52.3	17.3	56.4	15.5	57.4	19.9
S-COME (ICLR'25)	0.1	46.4	0.1	55.4	0.2	47.6	0.1	48.8	0.1	53.3	0.1	54.2	0.1	51.0	0.3
DPCore (ICML'25)	23.4	61.6	21.1	63.3	20.6	65.5	**25.0**	64.2	24.8	**55.6**	23.9	**59.8**	23.1	61.7	30.3
DOCO (Ours)	**28.1**	**62.2**	**24.4**	**64.8**	**25.7**	**78.2**	23.2	**71.2**	**26.1**	53.7	**24.6**	59.2	**25.4**	**64.9**	**32.7**

Table 3. Accuracy (%) under the closed-set setting.

Method	IN-C	IN-A	IN-R	IN-Sketch	IN-L	Avg.
Source	49.8	28.1	43.6	46.6	16.2	36.9
Tent [34]	58.1	29.8	44.1	46.6	0.84	35.9
CoTTA [36]	50.1	28.2	43.6	46.7	16.2	37.0
EATA [29]	59.9	29.9	46.0	47.6	23.4	41.4
OSTTA [20]	58.2	29.7	44.0	46.6	16.9	39.1
ViDA [24]	56.5	29.5	42.8	45.6	6.0	36.1
UniEnt [9]	59.1	29.3	45.4	47.6	22.5	40.8
STAMP [42]	51.7	28.8	43.6	46.7	16.2	37.4
E-COME [43]	59.0	31.2	46.3	49.2	15.5	40.2
DPCore [44]	**61.7**	29.3	46.3	49.0	24.5	42.2
DOCO (Ours)	60.6	**31.7**	**47.4**	**49.5**	**26.3**	**43.1**

Table 4. Ablation results for DOCO's modules.

Method	**Module**			**N (Norm-based)**		**P (DOCO)**	
	S	**O**	**R**	**H-score↑**	**Gain↑**	**H-score↑**	**Gain↑**
Source	-	-	-	56.4	-	56.4	-
w/o S.O.R	-	-	-	60.7	+4.3	64.0	+7.6
w/o S.O	-	-	✓	62.5	+6.1	67.6	+11.2
w/o O.R	✓	-	-	62.5	+6.1	65.1	+8.7
w/o O	✓	-	✓	63.9	+7.5	65.9	+9.5
w/o R	✓	✓	-	64.2	+7.8	68.5	+12.1
Full Model	✓	✓	✓	**68.1**	+11.7	**70.1**	+13.7

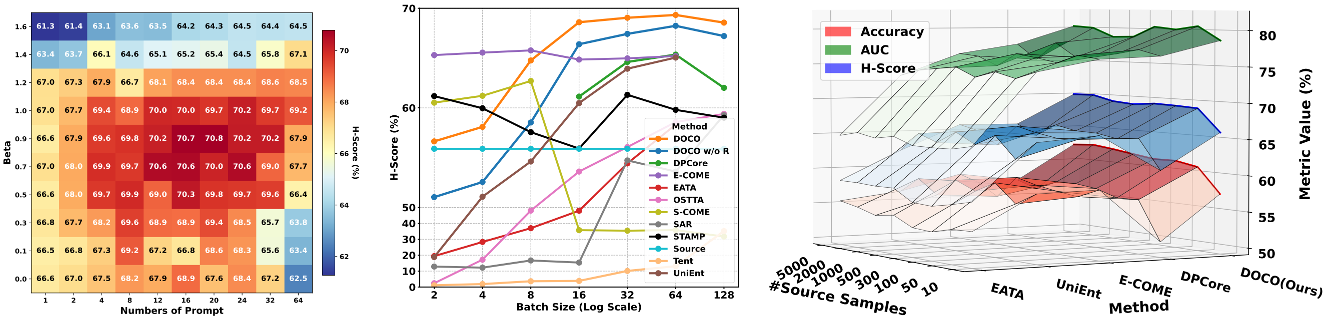
late the ratio ranging from 10% to 50%, and present the corresponding Accuracy and H-scores. The outcomes are averaged over six OOD datasets, with transparent bands indicating the standard deviation. While most baselines suffer from a significant performance drop and huge fluctuations as the ratios and datasets vary, DOCO shows the consistently superior and stable performance all the time.

LAION-C Benchmark. To explore the limits of adaptation under more extreme domain shifts, we evaluate our method on the recently released LAION-C benchmark [22]. As shown in Tab. 2, the severity of these shifts causes a drastic performance degradation across all methods, highlighting the benchmark's difficulty. Even in this adversarial setting, DOCO once again demonstrates its superior robustness and establishes a new state-of-the-art. Our method achieves the highest average H-score of **32.7**%, leading the closest competitor by **2.4**%. For dataset details and low-severity experiments, please refer to Appendix B.1 and C.3.

Closed-set CTTA. Additionally, we examine the methods under the fundamental closed-set scenario in Tab. 3. Apart from ImageNet-C and LAION-C, we introduce ImageNet-A [14], ImageNet-R [13] and ImageNet-Sketch [35] datasets, which provide natural adversarial, rendition and sketch domain shift. DOCO shows the best results on almost all datasets, highlighting its capability to handle a broad range of domain shifts, while also demonstrating that the design of DOCO does not harm performance in non-open-set scenarios. This outcome makes DOCO a potent competitor in the vanilla TTA and CTTA arena.

5.3. Analysis

Components Ablation Study. We conduct a comprehensive ablation study to dissect the contributions of DOCO's core components: Sample Splitting (**S**), OOD Propagation (**O**), and the Structural Regularizer (**R**). The results in Tab. 4 validate that our proposed mechanisms offer broad benefits, enhancing not only the prompt-based (**P**) approach but also the NormLayer-based (**N**) baseline, indicating the applicability to the ResNet backbone. A vanilla implementation that relies solely on statistical alignment (P w/o S.O.R) is insufficient, yielding a limited performance gain of +7.6%. DOCO's performance is achieved by building upon this foundation with its unique components. The synergistic application of **Sample Splitting (S)** and **OOD Propagation (O)** mechanisms propels the gain substantially to **+12.1**%. The final addition of the **Structural Regularizer (R)** further refines the feature space, culminating in the full DOCO model's H-score of 70.1%. Our analysis makes it clear that



(a) Sensitivity of β vs L (b) Batch size comparison (c) # Source Sample

Figure 7. (a) The parameter sensitivity analysis of β vs L on H-score (%). (b) The sensitivity of batch size on H-score (%). (c) Comparison of five methods with preliminary source characteristics extraction on three metrics versus the number of unlabeled source samples.

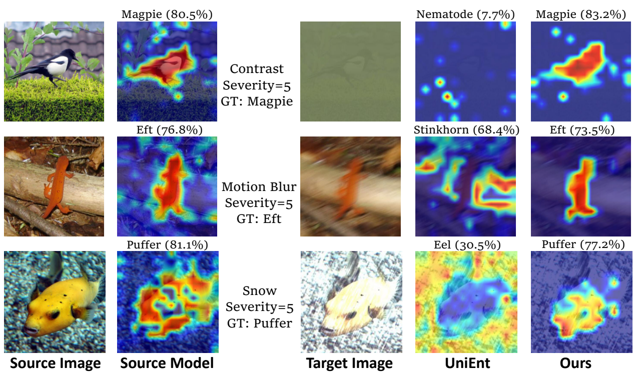


Figure 8. Grad-CAM visualizations for the source model on clean images and for UniEnt and DOCO on their corrupted counterparts.

DOCO's success is not merely a consequence of its adaptation paradigm, but is fundamentally driven by its specialized components, which are essential for unlocking robust performance in the demanding OCTTA setting. Notably, even when we disable the splitting mechanism (w/o S.O), the prompt still encodes a reasonable domain pattern on the mixed, noisy ID/OOD batch. This can be attributed to the structural regularizer, which prevents the prompt from aligning OOD semantics to source ID (w/o S.O.R $\rightarrow$ w/o S.O. yields an additional $+\mathbf{3.6}\%$). This further addresses a natural concern: even in extreme cases where we cannot clearly split the ID subset, the model does not collapse.

Sensitivity to prompt number L and regularization β. Figure 7a exhibits a broad plateau on the ImageNet-C and Places365-C combination: DOCO maintains high H-score across a wide swath of β and L, indicating low sensitivity to precise hyperparameter choices. Notably, along the diagonal trend where L increases, adaptation becomes harder to drive, as excess prompts diffuse learning and weaken domain compensation, but raising β counteracts this dispersion, preserving feature geometry and recovering accuracy.

Effect on batch size and source number. As shown in Fig. 7b, DOCO reaches first-tier performance beginning at batch size 8 and remains strong and stable thereafter. In the small-batch regime, removing the structural regularizer noticeably hurts performance relative to the full method, which matches the design insight in Sec. 4.1 when per-batch statistics are noisy. For source number in Fig. 7c, we compare five methods that leverage source number information. Across all source numbers, DOCO consistently attains the highest performance, evidencing high data efficiency. Remarkably, using only 50 source samples already delivers performance very close to the full setting. For details on the missing points and our small-batch solution of DOCO, please see Appendix B.3.

Visualization. As illustrated in Fig. 8, while the source model focuses on the target object in the clean domain, DOCO better recovers source-like attention under severe corruptions compared to UniEnt [9], concentrating on the true object regions rather than background noise. This qualitative comparison highlights DOCO's core capability: it disentangles domain shifts from semantic features, effectively compensating for the corruption and restoring source-like representations for reliable inference.

6. Conclusion

In this paper, we introduce OCTTA, a challenging yet pragmatic setting involving coupled domain and semantic shifts. To address this, we propose DOCO, a novel and efficient framework based on prompt learning. DOCO works by learning to compensate for domain shifts using likely ID data and then propagating this knowledge to enhance the detection of OOD samples, all within a self-reinforcing virtuous cycle. This synergistic design allows DOCO to remain robust and computationally practical where other methods falter. Extensive experiments on multiple challenging benchmarks validate the superiority of our method, which sets a new SOTA in handling coupled shifts.

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